

50.000 Papers



50.000 Papers



Planetary Boundaries



Climate Change



Stratospheric
Ozone



Biosphere
Integrity



Freshwater
Use



Biogeochemical
Flows



Atmospheric
Aerosol



Novel Entities



Ocean
Acidification



Land-System
Change

50.000 Papers

Planetary Boundaries

 Climate Change Stratospheric
Ozone

100 days straight without sleep!!!

ies

 Ocean
Acidification Land-System
Change

UPV-EARTH

Planetary Boundaries Framework for Classifying UPV Papers

AUTHORS

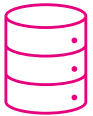
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OBJECTIVES

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Build the foundations of an AI system capable of mapping UPV research to the **Planetary Boundaries** framework.

1



Build a clean analytical corpus

Collect, clean and analyze UPV publications to ensure quality and consistency.

2



Define the PB classification task

Establish the 9 Planetary Boundaries, reference matrix, and evaluation criteria.

3



Compare NLP / LLM models

Evaluate embedding-based and LLM-based approaches to select the best solution.

4



Create a deployment prototype

Develop a functional interface to demonstrate real institutional usability.



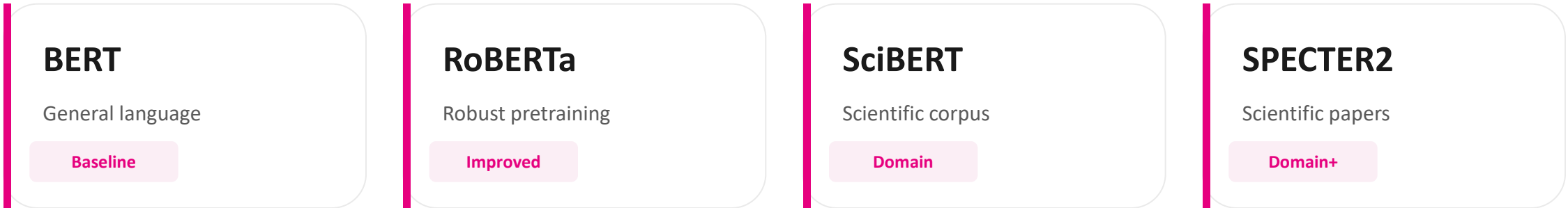
Goal:

Deliver a robust, evaluated and deployable prototype that transforms how UPV understands its contribution to the **Planetary Boundaries**.

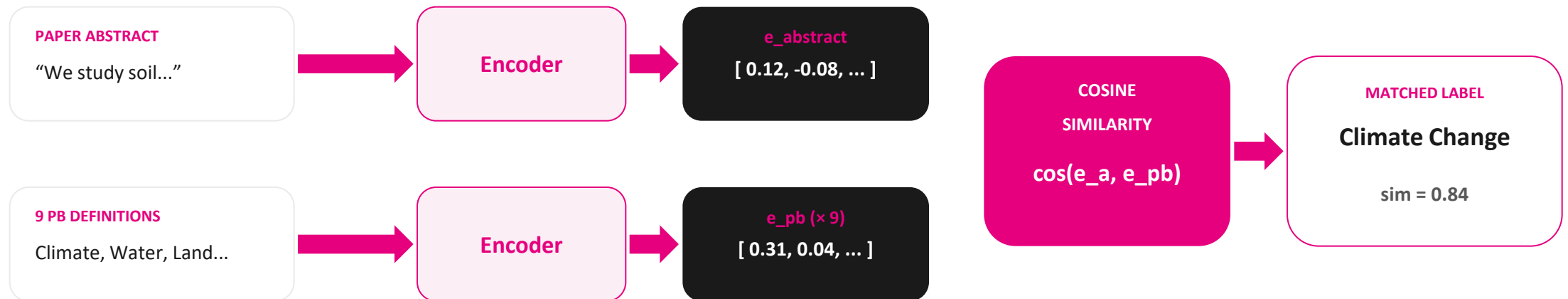
METHODOLOGY

TRANSFORMERS

Four pre-trained models compared, no fine-tuning required



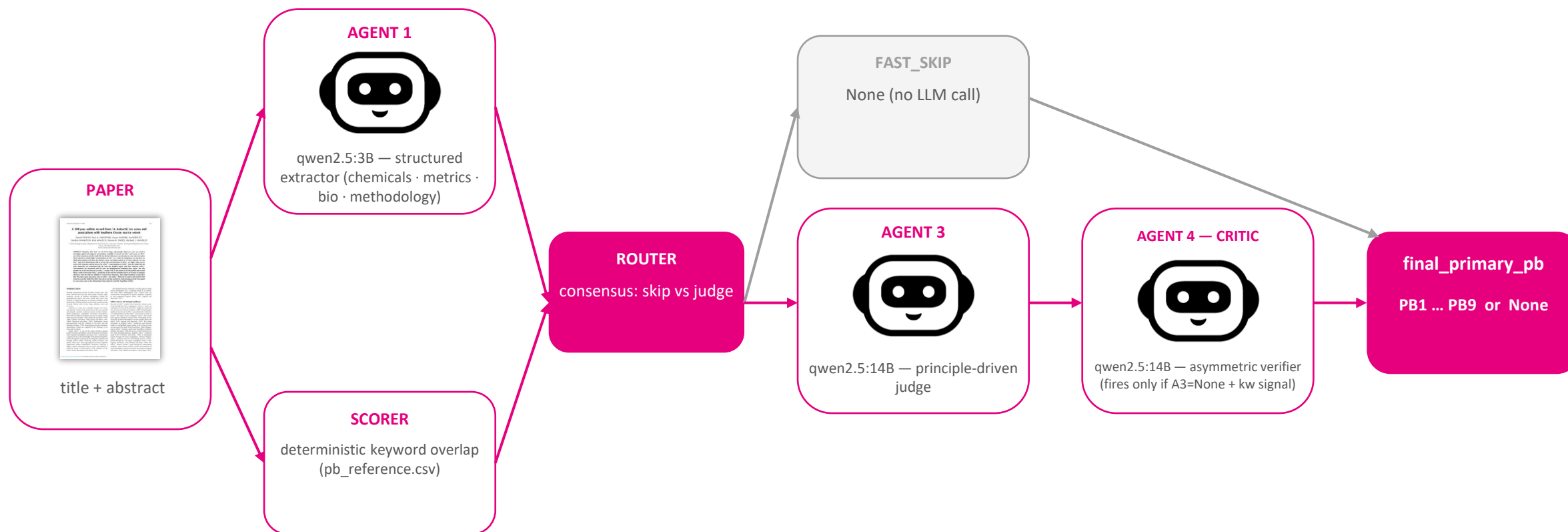
HOW WE PREDICT THE LABELS



Each abstract is encoded once, compared to all 9 PB embeddings, and assigned the boundary(ies) with highest cosine similarity.

AGENTS

A cascade of specialised LLM agents — light extractor, deterministic scorer, router, principled judge, asymmetric critic.



A lightweight first pass routes obvious cases to fast_skip; only ambiguous papers reach the heavier 14B judge and critic.

LLM MODELS

From candidate models to a principle-driven prompt — what we tried, what we kept.

MODELS EVALUATED

Llama 3.1 — 8B

Lightweight, fast — too shallow for nuanced PB reasoning.

Gemma 4 — 26B

Strong language fluency, weaker structured-output discipline.

Qwen 2.5 — 14B

Best trade-off: reasoning quality, instruction following, JSON/XML compliance.

✓ CHOSEN

PROMPT ENGINEERING — 4 APPROACHES

v1

Few-Shot

3–5 worked examples; brittle on unseen domains.

v2

CoT + Deductive Reasoning

Step-by-step justification before label; better recall.

v3

v2 + Background Filter

Discard generic intro sentences; focus on the contribution.

v4

Principle-Driven XML

Explicit PB principles; structured XML output; final choice.

✓ FINAL

Chosen pairing: Qwen 2.5 14B + principle-driven XML prompt (v4).

EVALUATION METRICS — ACCURACY

Six complementary metrics to measure how well models recover the correct planetary boundaries.

1 STRICT TOP-1

Only the single highest-scored prediction must exactly match the gold label.

2 RELAXED TOP-1

Top-1 prediction is correct if it belongs to any of the gold labels.

3 HIT@K

Any of the top-K predictions matches a gold label (K = 3 in our setup).

4 ANY OVERLAP

Prediction set intersects the gold set on at least one boundary.

5 EXACT MATCH

Prediction set equals the gold set — strictest multi-label metric.

6 JACCARD

$|pred \cap gold| / |pred \cup gold|$ — graded overlap between sets.

EVALUATION METRICS — BIAS AND BEHAVIOUR

Beyond accuracy: detecting model bias and characterising prediction behaviour.

BIAS DETECTION

TRUE NEGATIVE RATE (TNR)

Fraction of true negatives correctly identified as 'none'. Counters over-prediction.

POSITIVITY BIAS

Tendency to predict a label when none applies — surfaces over-confidence.

FALSE NEGATIVES

Missed gold labels — surfaces under-confidence and under-coverage.

BEHAVIORAL METRICS

AVG PBs / DOCUMENT

≈ 1.8

How many boundaries the model assigns per paper on average.

INFERENCE TIME

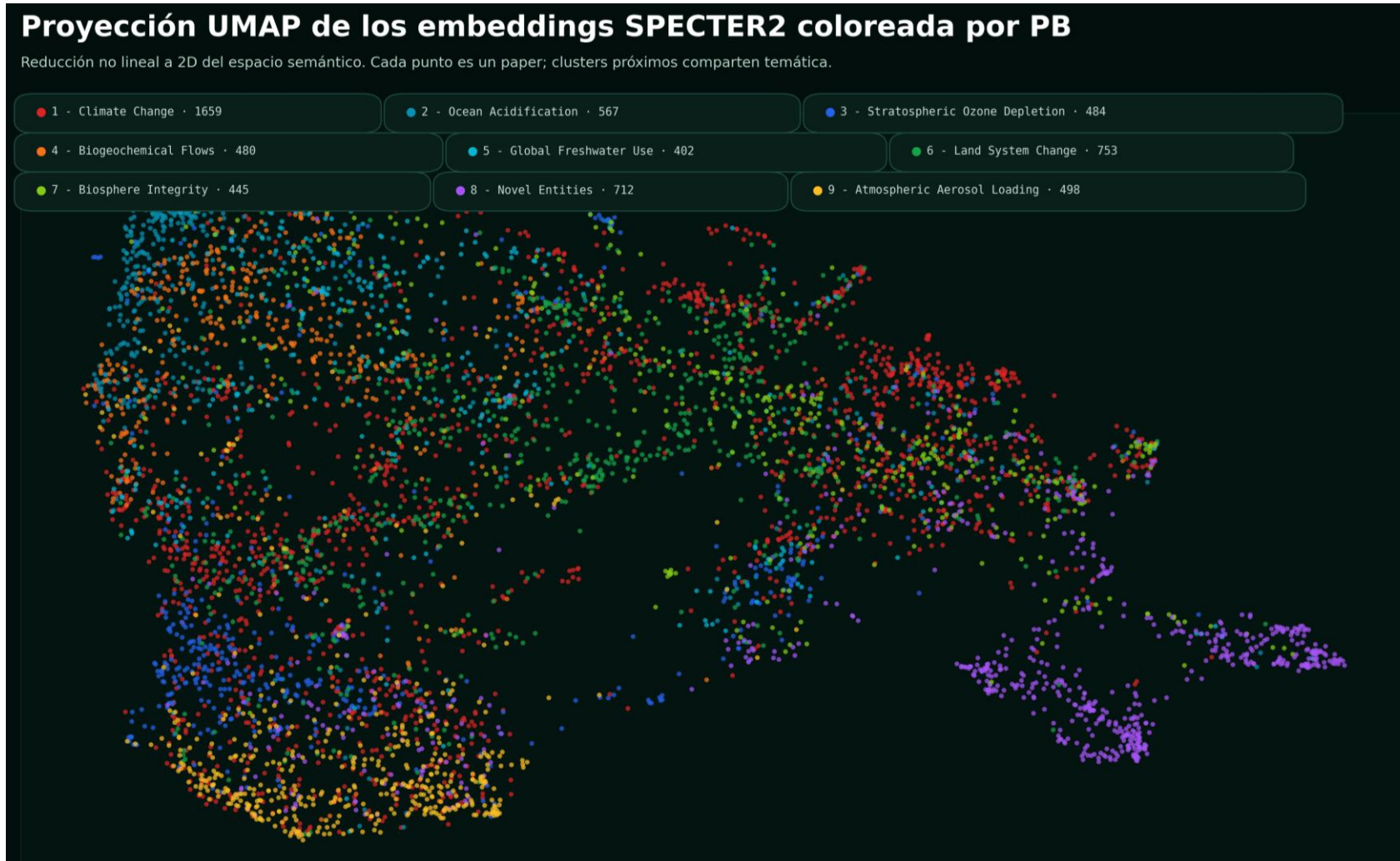
≈ 4 s/doc

Wall-clock latency per document on the cascade pipeline.

Together, accuracy + bias + behavior give a fuller picture than any single score.

RESULTS

EMBEDDINGS REPRESENTATION

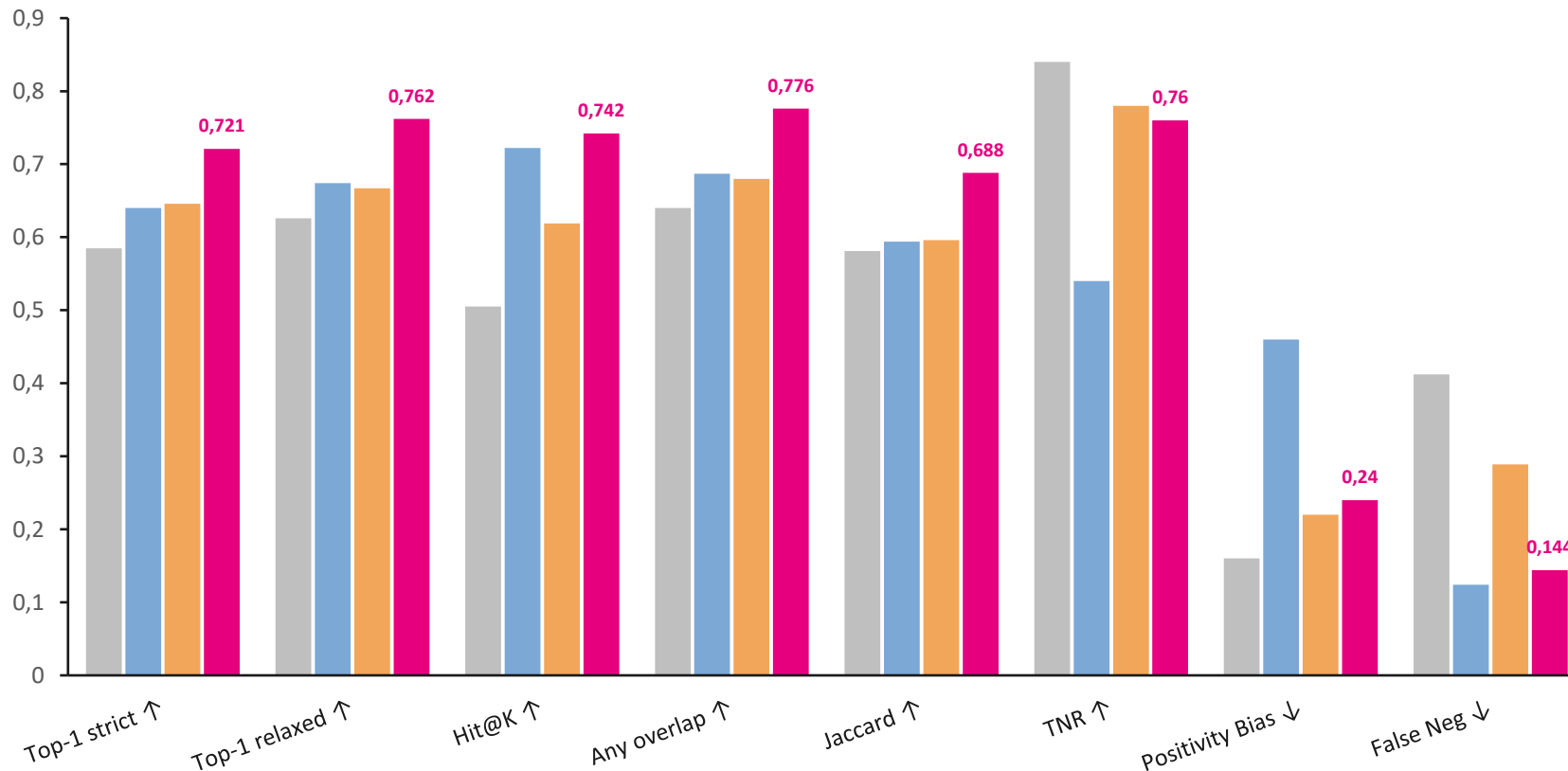


GLOBAL PERFORMANCE COMPARATION

Key metrics across the 4 prompt versions — v4 (Principle-Driven) wins on accuracy without losing bias control.

Metric scores by prompt version

■ v1 ■ v2 ■ v3 ■ v4 (final)



v4 PRINCIPLE-DRIVEN

Why it wins

+13.6 pts

Top-1 strict vs v1 (58.5 → 72.1%)

+13.5 pts

Any overlap vs v1 (64.0 → 77.5%)

+10.7 pts

Jaccard vs v1 (58.1 → 68.8%)

-26.8 pts

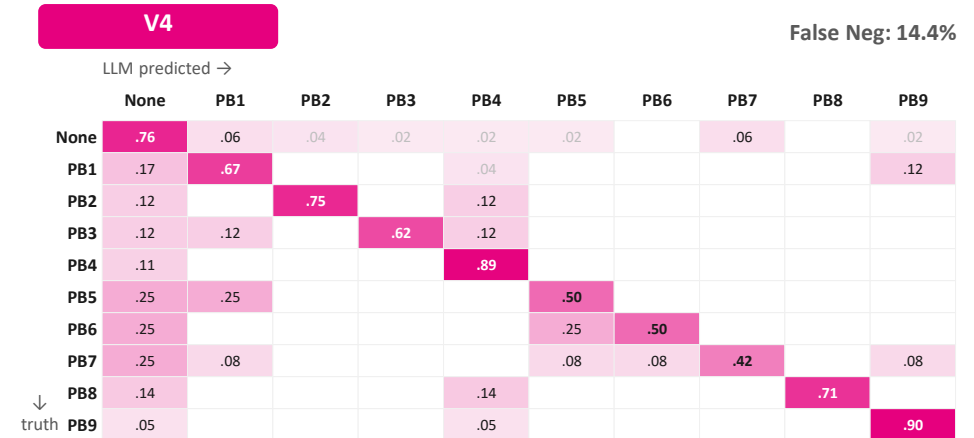
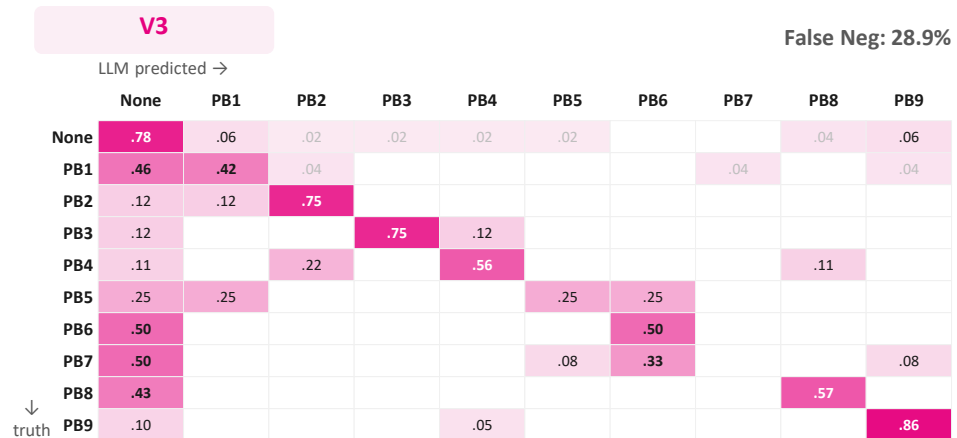
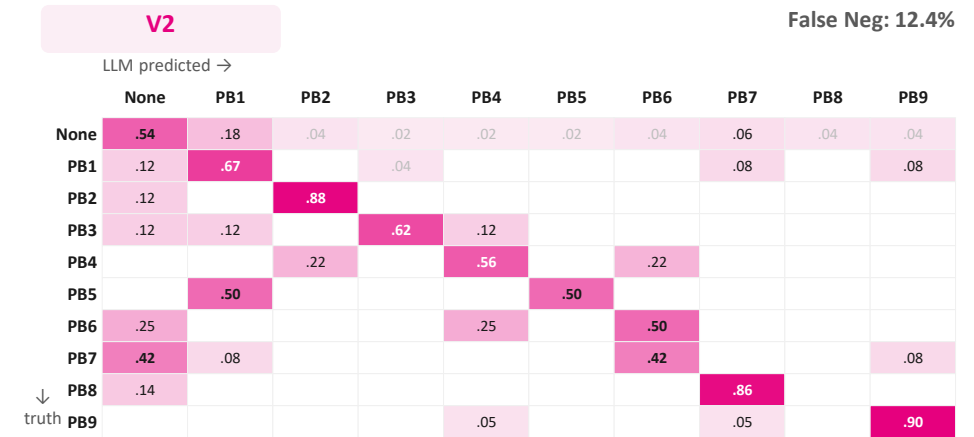
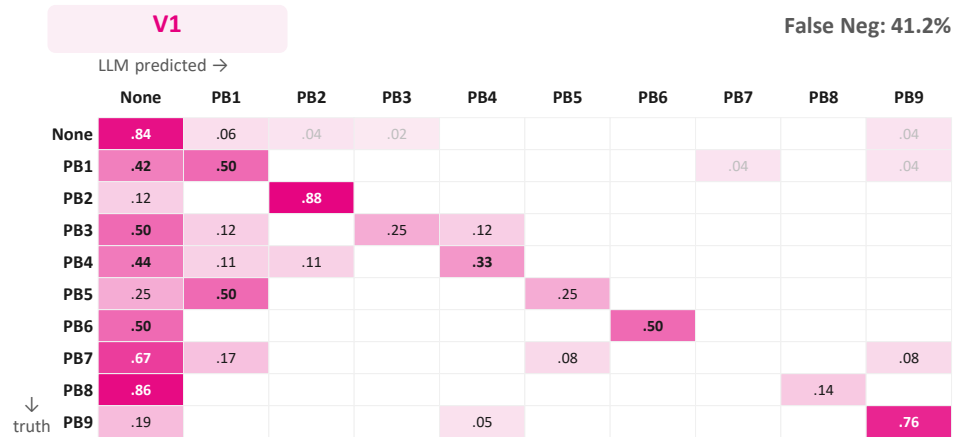
False Negatives vs v1 (41.2 → 14.4%)

TNR 76 %

Bias control (vs v2's 54 %)

CONFUSION MATRIX

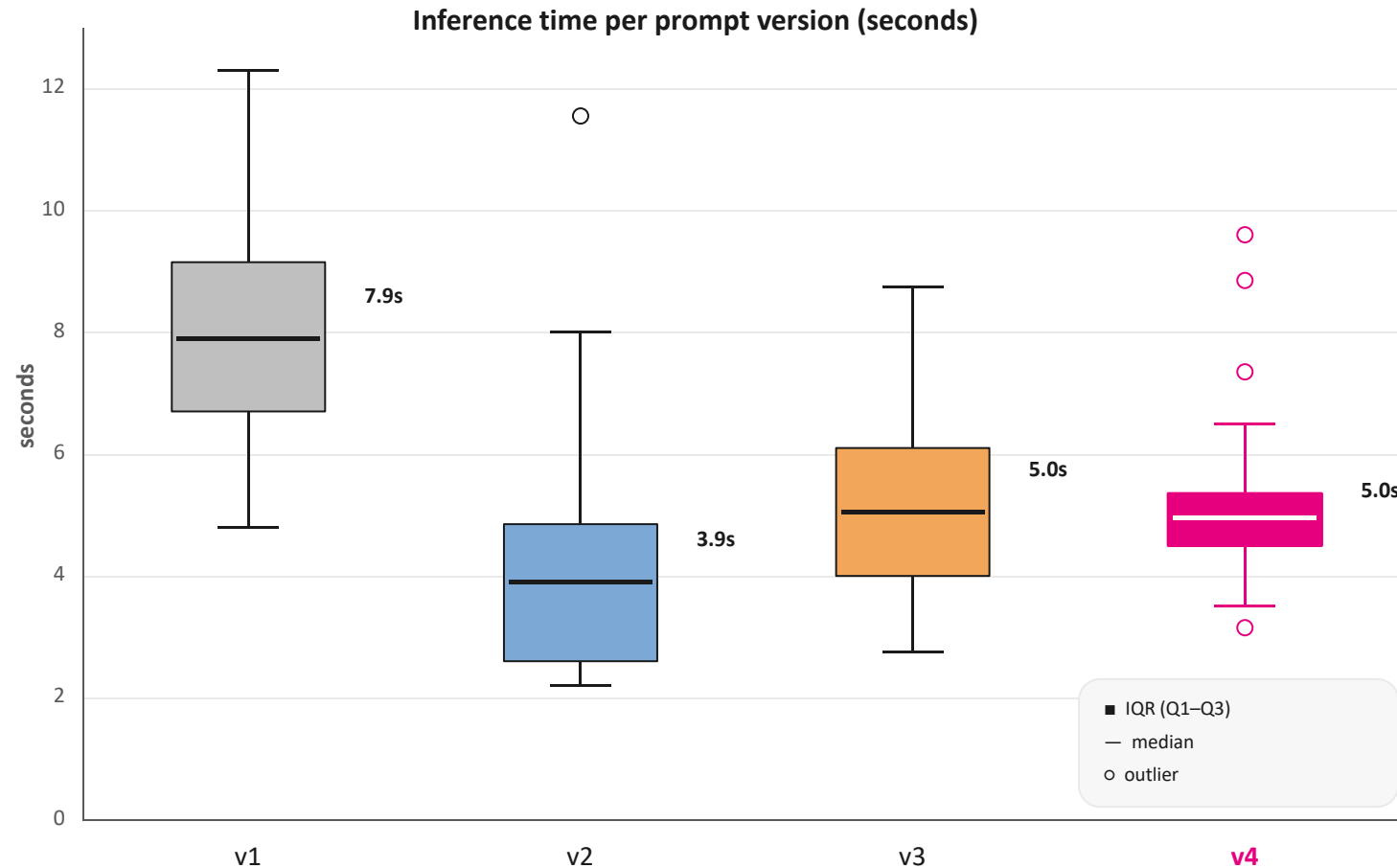
Normalized matrices (row = human label, column = LLM prediction). Stronger diagonal = better quality. v4 cleans up the diagonal across PB1–PB9.



v4 strengthens the diagonal on the hardest classes (PB4 +56 pts vs v1) while cutting false negatives by ~27 points.

INFERENCE TIME

Per-paper latency by prompt version — v4 is stable around 5 s/doc, well under v1's median 8 s.



LATENCY DISTRIBUTION

v1

med 7.9 s

IQR 6.7–9.2 s

min 4.8 · max 12.3 s

v2

med 3.9 s

IQR 2.6–4.9 s

min 2.2 · max 8.0 s

v3

med 5.1 s

IQR 4.0–6.1 s

min 2.7 · max 8.8 s

v4 (final)

med 4.7 s

IQR 4.4–5.3 s

min 3.5 · max 6.5 s

Verdict: v4 is the tightest distribution (IQR 4.4–5.3 s) and the fastest reliable choice — accuracy gains come with no latency penalty.

CONCLUSIONS AND LIMITATIONS

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CONCLUSIONS

Mastering Behavioral Divergence



Prompting = Reasoning Design

Success depends on aligning instructions with the model's internal logic.



Behavioral Divergence

XML vs. Few-Shot prompts drastically change how the model interprets boundaries.



Winning Formula

A principle-driven approach balances rigor (True Negatives) with coverage (Recall).

LIMITATIONS

The Data Gap



Validated Paper Scarcity

Lack of human-labeled ground truth limits statistical significance.



Fine-Tuning Barrier

Insufficient data prevents supervised fine-tuning of Transformers and LLMs.



Rare Class Uncertainty

Low sample sizes for PB5 and PB6 hinder performance validation.